Is Istanbul Running Out of Water? Drought, Climate Change and the City's Reservoirs

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Project Summary

Istanbul is home to more than 15 million people, yet almost all of its drinking water comes from a small group of reservoirs that depend on rainfall. In recent years these reservoirs have repeatedly dropped to alarming levels; in October 2025 the city-wide average fell below 25%, and some dams were nearly empty. This project will explore how Istanbul's dam occupancy has changed over the last two decades and how these changes relate to rising temperatures and shifting rainfall patterns in the region. Using several types of visualizations (time series, seasonal heatmaps, comparisons between individual dams and climate anomaly charts), the goal is to show that climate change is not a distant problem but something that directly affects the water we use every day, connecting the topic to SDG 13 (Climate Action) and SDG 6 (Clean Water and Sanitation).

Dataset Information

1. Istanbul Dam Occupancy Rates

- Data Source: Istanbul Metropolitan Municipality (IBB) Open Data Portal
- Original Data Provider: Istanbul Water and Sewerage Administration (ISKI)
- Content: Daily occupancy rates (%) of Istanbul's reservoirs, both city-wide and for each individual dam
- Dataset Links:
  - City-wide: https://data.ibb.gov.tr/dataset/istanbul-dam-occupany-rates-data
  - Per dam (daily): https://data.ibb.gov.tr/dataset/istanbul-barajlari-gunluk-doluluk-oranlari

2. ERA5 Monthly Reanalysis (Temperature and Precipitation)

- Data Source: Copernicus Climate Change Service (C3S), Climate Data Store
- Content: Monthly 2 m air temperature and total precipitation for the Istanbul region
- Dataset Link: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels-monthly-means

3. Climate Change Knowledge Portal – Türkiye

- Data Source: World Bank Group
- Content: Historical climate trends and future projections (temperature, precipitation, drought indicators) for Türkiye
- Dataset Link: https://climateknowledgeportal.worldbank.org/country/turkiye